The Game Beyond the Scoreline: Uncovering the Value of Football Actions Through VAEP-Based Analytics and Interactive Exploration

Shreyank Parab (24102B0044), Malhar Kamble (24102B0050), Amaan Khan (24102B0036)
Mentor - Prof. Mahesh Khandke
Department of Computer Engineering
Vidyalankar Institute of Technology, Mumbai, India

***Abstract*—This paper presents *VAEP Explorer*, an end-to-end football event analytics system that computes and visualizes action-level VAEP. StatsBomb Open Data for all 64 FIFA World Cup 2022 matches is converted to SPADL; two CatBoost classifiers estimate short-horizon scoring and conceding probabilities, whose state-to-state changes form offensive, defensive, and total VAEP. Goal and possession facts are separately joined to the existing action records to support goal-centered replay without changing model values. The resulting 138,946 action records, including 41 zero-VAEP penalty-shootout events, are stored in Supabase PostgreSQL and served by an Express API to a React application. We evaluate fresh XGBoost and CatBoost fits using five repeated match-grouped holdouts, report threshold and probability-quality metrics against a training-prevalence baseline, and separately audit stored data integrity. At a fixed threshold of 0.50, scoring recall is 0.1180 for CatBoost, while conceding recall is 0.0000 for both fitted models despite high accuracy under severe class imbalance. These results support an exploratory system contribution, not claims of dependable rare-event detection or cross-competition generalization.**

***Index Terms*—football analytics, VAEP, SPADL, StatsBomb, CatBoost, match-grouped validation, action valuation, interactive visualization**

I. Introduction

Football performance analysis has historically concentrated on outcomes that are immediately visible in the scoreline, particularly goals and assists. Such summaries are valuable, but they compress a long sequence of contextual decisions into a small number of terminal events. Passes, carries, recoveries, tackles, and other actions can change the probability of a favorable future state well before a shot or goal occurs. VAEP (Valuing Actions by Estimating Probabilities) formalizes this intuition by estimating how an action changes the short-term probability of scoring and conceding [1].

The system covers the complete path from raw event records to stored analytical output. Events are standardized, contextual features and future-outcome labels are generated, model scores are converted into action values, and the resulting records are persisted before runtime analysis.

The system has three objectives: to process public football event data through the VAEP workflow, to keep model construction outside the runtime request path, and to support match- and player-level exploration of the resulting action values.

The evaluation focuses on reproducible system implementation and a transparent within-tournament validation experiment, not production-level scouting performance. The final stored VAEP seed covers all 64 tournament matches; classifier evaluation holds out complete matches to avoid splitting actions from one match across train and test.

II. Literature Review / Related Work

Football analytics has evolved from traditional statistics, such as goals, assists, and possession percentages, toward more context-aware methods that evaluate the contribution of individual actions. Existing research provides the theoretical foundations for standardized football event representation, action valuation, predictive modelling, and interactive visual analysis. The following studies are particularly relevant to the design and implementation of *VAEP Explorer*.

A. Action Valuation in Football

Decroos *et al.* introduced SPADL as a standardized language for football actions and proposed VAEP for valuing actions according to their contextual effect on subsequent scoring and conceding probabilities [1]. This approach is particularly useful for a system such as *VAEP Explorer*because it converts heterogeneous event logs into a common analytical vocabulary and assigns value to actions that do not necessarily end in goals or shots.

Action-value modelling is part of a broader family of possession-value approaches. Fernández *et al.* developed an expected possession value framework that models the instantaneous value of a possession and decomposes contributions from passes, carries, and shots using tracking information [4]. Expected-threat-style approaches similarly assign value to locations and transitions, illustrating the design spectrum between spatial state-based models and richer action-context models. Recent research continues to

investigate how timing and possession dynamics can be incorporated into these metrics, reinforcing the importance of temporal context in football valuation [5].

B. Public Event Data and Football Analytics Systems

The availability of structured football event datasets has enabled reproducible academic research that was previously constrained by proprietary data. Pappalardo *et al.* described an open collection of spatio-temporal football events containing information such as action type, timestamp, player, location, and outcome [2]. StatsBomb Open Data extends this ecosystem through openly distributed competition, match, event, and lineup files that can be processed locally [8].

Visual analytics has also been applied to football to support the exploration of player performance, match events, and tactical patterns. Janetzko *et al.* developed feature-driven visual views for examining player movement and game events [10], while Delibas *et al.* presented an interactive exploratory system combining analytical models with visualization and replay [11]. SoccerNet provides a complementary video-based perspective through action spotting in football footage [3]. In contrast, the present system focuses on event-data-based action valuation and connects model-derived values with database-backed, interactive match exploration.

C. Research Gap and Contribution

Although previous studies establish standardized action representation, contextual valuation, and visual exploration as important components of football analytics, integrating these elements into a reproducible end-to-end application remains a practical engineering challenge. In particular, model evaluation, stored-value integrity, and interactive match exploration must be addressed together without confusing predictive performance with the usefulness of the resulting analytical interface.

Against this background, the present work makes the following contributions:

- **Full-tournament implementation:** Applies the SPADL and VAEP workflow to the complete StatsBomb event dataset for the 2022 FIFA World Cup, covering all 64 matches.
- **Separated model evaluation and deployment:** Evaluates fresh XGBoost and CatBoost classifiers using match-grouped holdouts, while separately fitting the full-data CatBoost models used to generate the stored VAEP values.
- **Database-backed analytical exploration:** Integrates prepared action values into a persistent database and an interactive application supporting match replay, goal-centered exploration, and player-level comparisons.
- **Data-integrity and performance reporting:** Audits stored records and reports rare-event classification limitations, ensuring that high accuracy under class imbalance is not presented as evidence of dependable event detection.

D. Conceptual Foundations: Event Data, SPADL, and VAEP

Understanding the proposed system requires distinguishing the underlying event data from the standardized action representation and the subsequent valuation process.

StatsBomb Open Data supplies event-stream records: annotated on-ball events with event-specific details such as time, location, type, outcome, and player, together with match and lineup information [8]. It is not optical tracking data; it does not provide continuous trajectories for every player and the ball. Consequently, the analysis is limited to recorded events and their locations rather than off-ball movement or continuous physical tracking.

SPADL (Soccer Player Action Description Language) standardizes provider events into a sequence of interpretable player actions [1], [9]. An action is a player-performed event on the ball; match events that are not player actions are not treated as equivalent action records. In this workflow, the StatsBomb converter produces actions with a common set of fields, including action type, result, body part, start and end locations, time, player, and team. This standardization allows the same feature-generation and valuation procedures to operate across passes, carries, shots, defensive interventions, and other converted actions rather than assigning value only to goals or shots.

VAEP does not assign a fixed bonus to an action label. Instead, it estimates the action's contextual effect on two short-horizon outcomes: whether the acting team scores and whether it concedes. For example, a completed forward pass can be valuable when it improves the next scoring state, while a similar pass under different pressure, location, or game context can have a different value. A tackle, interception, or clearance can contribute defensively when it lowers the estimated probability of conceding. A shot is not automatically positive: its value is determined by the before-to-after probability changes, not simply by the action name or whether a goal eventually appears in the match.

Table I presents illustrative action groups and explains how their contextual effects are represented within VAEP. These are not separate hand-authored scoring rules; both classifiers evaluate included standard-play actions using the same contextual modelling framework.

III. Proposed Methodology / Architecture

A. Research Design

The study combines computational systems design with an exploratory predictive evaluation. Its unit of analysis is a standard-play action represented by the action and three preceding actions. Two binary classifiers estimate whether the acting team scores or concedes within a ten-action horizon. Their probabilities yield offensive and defensive VAEP components. We assess predictive performance on

TABLE I
ILLUSTRATIVE SPADL ACTION GROUPS AND THEIR ROLE IN VAEP.

Action group	**How context can affect its value**
Ball progression and retention (e.g., passes, crosses, carries)	Location, result, and subsequent state can raise or lower the acting team's scoring probability.
Attempts (e.g., shots)	The change in both scoring and conceding probabilities is valued; an attempt is not assigned a fixed positive value.
Defensive interventions (e.g., tackles, interceptions, clearances)	Actions can add defensive value when they reduce the acting team's estimated probability of conceding.
Restarts and goalkeeper actions	Their value depends on the game state and resulting probabilities, not on a universal action-type constant.

held-out matches and separately verify the integrity of generated and stored records. These checks do not establish tactical correctness or causal attribution.

B. Data / Participants

The experiment uses all 64 FIFA World Cup 2022 matches in StatsBomb Open Data (competition 43, season 106) [8]. The resulting seed contains 680 players, 138,946 action records, and 680 player-stat records. Of the action records, 138,905 are standard-play actions and 41 are penalty-shootout actions. Shootout actions remain available in replay but are excluded from model fitting, validation, and player VAEP aggregates; their VAEP components are zero.

Regular-play goal annotations identify 169 scored shots and three own goals (172 goals total), with credited score-after fields reconciled against all match final scores.

Each observation is derived from football event and lineup records. The data are downloaded and cached locally, and the deployed application does not consume a live match feed.

C. Materials and Tools

The offline stage uses StatsBomb event files and the socceraction toolkit for event conversion, SPADL generation, and VAEP-oriented feature and label construction [8], [9]. CatBoost is used to generate the deployed VAEP values; XGBoost is an additional comparator in held-out validation [6], [7]. The application layer uses Supabase-hosted PostgreSQL, Drizzle ORM, Zod, Express, and React for storage, validation, API delivery, and client-side visualization.

D. Procedure

The implementation is organized into offline preparation and online serving. The offline stages are summarized in Fig. 1, while the deployed request path is shown in Fig. 2.

First, the downloader acquires the competition catalog, the selected competition-season matches, and the events and lineups for the chosen matches. Files are cached locally unless overwrite is explicitly requested. Each game is loaded through the StatsBomb loader and converted to SPADL actions. The actions are then normalized to a left-to-right attacking orientation so that pitch coordinates retain a consistent analytical interpretation across halves.

Each model input is built from a four-action window containing the target action and its three predecessors. The representation combines action type, result, body part and time; start/end locations and goal-relative polar coordinates; movement; team and score state; and temporal and spatial differences [9]. The field orientation is normalized to left-to-right. CatBoost receives categorical features as categories and XGBoost receives numeric category codes; missing and non-finite inputs are handled during preprocessing.

The labels are generated from future outcomes, not supplied as features. Let g_j^s and g_j^c indicate whether the acting team scores or concedes on action j. For standard-play action index i within a match, the targets mark whether either event occurs during the ten-action window beginning at i, including the current action and the next nine:

$$y_i^s = \mathbf{1}\left[\sum_{j=i}^{i+9} g_j^s > 0\right], \qquad y_i^c = \mathbf{1}\left[\sum_{j=i}^{i+9} g_j^c > 0\right]. \tag{1}$$

The scoring and conceding tasks are learned separately from the same action context. A positive conceding label means the opponent scores within the horizon. Shootout-period actions are removed before feature and label generation for both validation and final model training.

For validation, `GroupShuffleSplit` creates five random holdouts at the match level. Each split uses 48 training matches and 16 test matches (25% test fraction), with seeds 42–46. All actions belonging to a match remain together. XGBoost and CatBoost are fit independently on identical partitions for each target; both use 100 boosting iterations, depth 5, and learning rate 0.05. XGBoost additionally uses row and column subsampling of 0.8 and a binary-logistic objective. CatBoost uses Logloss and native categorical features.

A constant predictor equal to the training-set positive prevalence is the baseline. Threshold-dependent metrics use 0.50. The validation report records XGBoost 2.1.4, CatBoost 1.2.8, scikit-learn 1.9.0, pandas 2.3.3, and NumPy 1.26.4. Because the five random test sets overlap, the reported mean and sample standard deviation describe split variability and are not confidence intervals.

The deployed seed is generated separately: CatBoost scoring and conceding models with 100 iterations, depth 5, learning rate 0.05, Logloss, and random state 42 are fitted on all standard-play actions from the 64 matches after the

validation experiment. Thus, the validation scores below are not scores of the final all-data models or predictions of the stored action values.

Let S_i denote the game state after action a_i, and let X_i be its model input: the features of a_i and its three preceding actions. The two models estimate:

$$\begin{aligned} P_s(S_i) &= \Pr(y_i^s = 1 \mid X_i), \\ P_c(S_i) &= \Pr(y_i^c = 1 \mid X_i). \end{aligned} \tag{2}$$

Thus, P_s and P_c are the acting team's model-estimated scoring and conceding probabilities for the defined ten-action outcome window, not goal counts. Action a_i moves the state from S_{i-1} to S_i, so its offensive and defensive contributions are:

$$\begin{aligned} V_{\text{off}}(a_i) &= P_s(S_i) - P_s(S_{i-1}), \\ V_{\text{def}}(a_i) &= P_c(S_{i-1}) - P_c(S_i), \\ V_{\text{VAEP}}(a_i) &= V_{\text{off}}(a_i) + V_{\text{def}}(a_i). \end{aligned} \tag{3}$$

The signs make both components positive for a favorable change: offensive value rises when scoring probability increases, and defensive value rises when conceding probability decreases. The components are added, not subtracted from one another; equivalently, total VAEP is the increase in scoring probability minus the increase in conceding probability.

For example, +0.05 is a five-percentage-point favorable change in model-implied net near-term probability, not a guaranteed goal or a literal goal count. A negative total can therefore indicate that an action worsened the estimated state, even if the player later scores or the team eventually wins. These probability differences are not calibrated estimates, causal effects, or standalone measures of player quality. Player totals in this implementation sum the stored action values over standard play, with a separate per-action average; they are not the per-90-minute player rating used in some prior studies. The models run offline, and their outputs are written with match, player, action, and player-summary records to a JSON seed.

Goal-story fields are added in a distinct enrichment step that does not train models or recalculate VAEP. Regular-play StatsBomb shots with a goal outcome and own-goal-against events are mapped to SPADL actions by source event ID; own goals are credited to the opposite side. Period-five shootout events are excluded. Each action also receives its StatsBomb possession identifier when available.

For generated actions without a source event ID, possession may be associated with a same-period, same-team, same-player event no more than eight seconds away; goal events themselves require an exact ID match. Goal counts and score-after progression are checked against each match's final score. In the resulting data, all 172 goals reconcile and all goal actions have possession IDs. Possession-bounded buildup is presented as recorded event context, not evidence that each action caused the goal.

The seed importer validates the four top-level arrays using Zod, maps external StatsBomb identifiers to internal PostgreSQL identifiers, inserts parent records before dependent action records, and writes actions and player summaries in batches of 500. Conflict-aware inserts update existing rows rather than creating duplicates. The entire import runs transactionally, so an invalid reference or failed record can roll back the operation. The resulting database is exposed to the application through read-only access policies.

At runtime, the browser requests the Express API, which queries the prepared Supabase data. The React client uses generated API clients and React Query to manage requests and loading/error states. The interface exposes a dashboard, sortable leaderboard, action replay, and quality-versus-quantity scatter view. Action replay retrieves a selected match by its external identifier and renders a pitch representation with arrows between stored start and end coordinates.

E. Data Analysis

The validation metrics are calculated independently for each held-out match group at threshold $\tau = 0.50$. For target $y_j \in \{0, 1\}$ and predicted probability p_j, the hard prediction is $\hat{y}_j = \mathbf{1}[p_j \geq \tau]$. Here, TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives.

Accuracy is $(TP+TN)/N$, precision is $TP/(TP+FP)$, recall is $TP/(TP+FN)$, and F_1 is their precision–recall harmonic mean; undefined precision, recall, or F_1 is reported as zero, matching the validation implementation. Balanced accuracy is the mean of sensitivity and specificity. ROC AUC measures ranking over false-positive rates, while PR AUC is average precision over the precision–recall curve; these ranking metrics are not tied to the fixed threshold.

Probability errors are measured by the Brier score and binary log loss:

$$\begin{aligned} \text{Brier} &= \frac{1}{N}\sum_{j=1}^{N}(p_j - y_j)^2, \\ \text{LogLoss} &= -\frac{1}{N}\sum_{j=1}^{N}\ell_j, \\ \ell_j &= y_j \log p_j + (1 - y_j)\log(1 - p_j). \end{aligned} \tag{4}$$

For each metric m, one score m_r is calculated per split. We report its arithmetic mean and sample standard deviation:

$$\begin{aligned} \bar{m} &= \frac{1}{R}\sum_{r=1}^{R} m_r, \\ s_m &= \sqrt{\frac{1}{R-1}\sum_{r=1}^{R}\delta_r^2}, \\ \delta_r &= m_r - \bar{m}, \qquad R = 5. \end{aligned} \tag{5}$$

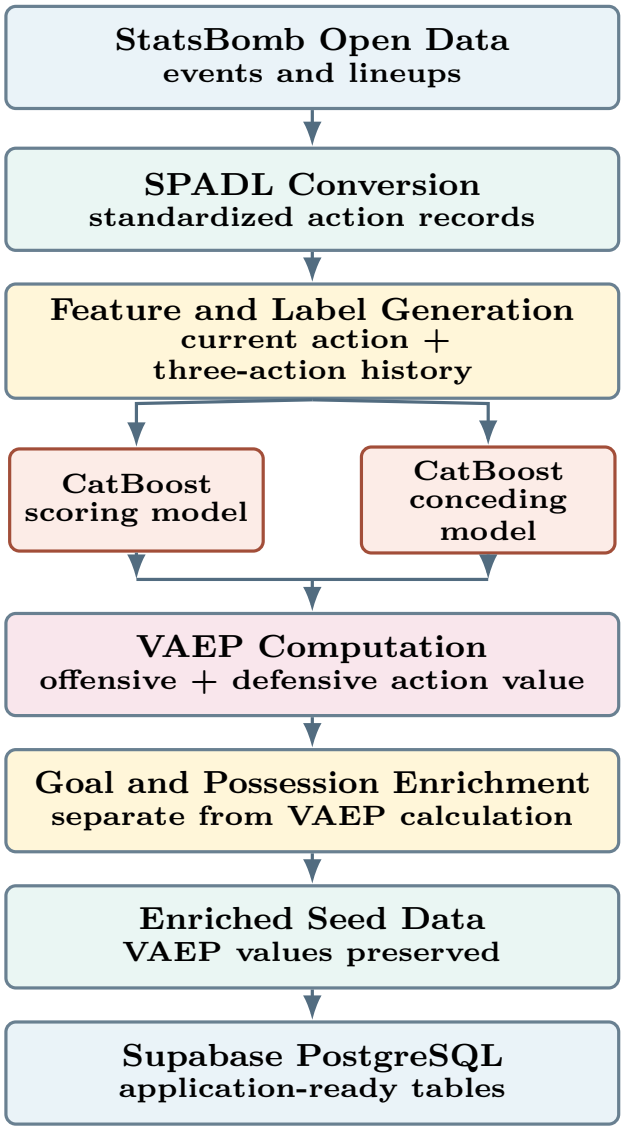


Fig. 1. Offline preparation workflow. StatsBomb event and lineup records are converted to SPADL actions, represented with contextual features and short-horizon labels, scored by separate CatBoost models, and transformed into VAEP values. Goal and possession facts are joined in a separate enrichment step before validated records are stored for application use.

These overlapping holdouts are repeated random assessments, not independent samples. The training-prevalence baseline assigns every test action the positive-class frequency observed in that split's training set; at threshold 0.50 it predicts no positive events because both target prevalences are below 0.50.

IV. Results

A. Implemented Data Scope and Integrity

The deployed pipeline covers all 64 FIFA World Cup 2022 matches. Table III summarizes its scope and internal verification. The seed includes 138,946 actions, of which 41 are shootout events kept for replay but assigned zero VAEP.

TABLE II
Feature and label groups used by the pipeline.

Group	Role in modelling
Action semantics	Type, result, body part, and match time describe the immediate event.
Spatial state	Start/end location, polar position relative to goal, and movement encode field context.
Temporal context	Temporal and spatial differences capture change around the current event.
Game state	Team state and score state provide match context at action time.
Short history	The current action plus the three preceding actions represent local game state.
Outcome labels	Scoring and conceding outcomes are derived over the current and next nine actions.

TABLE III
Dataset scope and stored-data verification.

Measure	Observed implementation result
Match scope	All 64 FIFA World Cup 2022 matches (StatsBomb competition 43, season 106).
Stored data	680 players; 138,946 actions; 680 player-stat rows.
Action split	138,905 standard-play actions; 41 shootout actions with zero VAEP.
Goal enrichment	172 regular-play goals (169 scored shots and 3 own goals); zero shootout goals; zero final-score mismatches.
Persistence	Enriched seed imported into Supabase; the goal-story migration extends the existing schema.
Arithmetic integrity	Zero VAEP component-sum or player-stat aggregation mismatches in the seed audit.
Database integrity	Schema and RLS/permission checks pass; no public table writes; valid coordinate bounds and finite VAEP values.

The goal-story enrichment attaches event-derived facts without changing the VAEP values or player aggregates.

B. Match-Grouped Predictive Validation

Table IV reports held-out scoring metrics. CatBoost attains slightly higher mean accuracy and recall than XGBoost, whereas XGBoost has a slightly higher mean ROC AUC and PR AUC. At the selected threshold, both models have 1.0000 precision but low recall, so the apparent accuracy does not indicate that most future scoring events are detected.

Table V reports held-out conceding metrics. Both fitted models predict no positive cases at threshold 0.50 in the reported splits, resulting in zero precision, recall, and F_1; their accuracy matches the rounded prevalence baseline. CatBoost has higher mean ROC AUC and PR AUC than XGBoost, but these ranking results do not remove the operating-threshold failure.

These are repeated holdout results for fresh fits, not a comparison of the final CatBoost seed model's predictions. The configurations are fixed rather than tuned, and the

TABLE IV
HELD-OUT SCORING OUTCOME METRICS (MEAN ± SAMPLE SD OVER FIVE MATCH-GROUPED SPLITS).

Metric	XGBoost	CatBoost	Training-prevalence baseline
Accuracy	.9905 ± .0011	.9906 ± .0010	.9893 ± .0011
Balanced accuracy	.5544 ± .0035	.5590 ± .0013	.5000 ± .0000
Precision	1.0000 ± .0000	1.0000 ± .0000	.0000 ± .0000
Recall	.1089 ± .0070	.1180 ± .0025	.0000 ± .0000
F_1	.1963 ± .0113	.2110 ± .0040	.0000 ± .0000
ROC AUC	.7352 ± .0149	.7249 ± .0171	.5000 ± .0000
PR AUC (average precision)	.1706 ± .0074	.1662 ± .0061	.0107 ± .0011
Brier score	.0094 ± .0010	.0093 ± .0010	.0106 ± .0011
Log loss	.0510 ± .0048	.0512 ± .0045	.0592 ± .0053

TABLE V
HELD-OUT CONCEDING OUTCOME METRICS (MEAN ± SAMPLE SD OVER FIVE MATCH-GROUPED SPLITS).

Metric	XGBoost	CatBoost	Training-prevalence baseline
Accuracy	.9980 ± .0004	.9980 ± .0004	.9980 ± .0004
Balanced accuracy	.5000 ± .0000	.5000 ± .0000	.5000 ± .0000
Precision	.0000 ± .0000	.0000 ± .0000	.0000 ± .0000
Recall	.0000 ± .0000	.0000 ± .0000	.0000 ± .0000
F_1	.0000 ± .0000	.0000 ± .0000	.0000 ± .0000
ROC AUC	.7307 ± .0307	.7561 ± .0257	.5000 ± .0000
PR AUC (average precision)	.0169 ± .0038	.0227 ± .0062	.0020 ± .0004
Brier score	.0020 ± .0004	.0020 ± .0004	.0020 ± .0004
Log loss	.0136 ± .0024	.0134 ± .0025	.0146 ± .0027

five test sets overlap. Published metrics from other studies are not directly comparable because their data, labels, and split protocols differ.

C. Application-Level Result

The browser-facing implementation provides a match report and replay, a goal-first match story, key moments, team activity, player comparisons, action sequences, a leaderboard, and a quality-versus-quantity view. In replay, goal markers are distinct from non-goal shots; goal cards show scorer, time, credited team, and score after the goal. Selecting a goal displays the pitch event and up to ten recorded actions from the same possession when available. Shootout actions are displayed separately and excluded from VAEP.

This presentation targets both analysts and casual viewers while retaining the warning that possession context is not causal evidence. The runtime path is read-oriented: the client does not trigger new downloads, model training, or VAEP calculations.

D. Reproducibility and Validation

Reproducibility is supported by explicit pipeline parameters: competition identifier, season identifier, match limit, overwrite behavior, output path, train/holdout fraction, and random state. Re-running the same workflow over the same cached inputs reproduces the same selection and split. External StatsBomb identifiers are retained as interchange keys while PostgreSQL uses internal identifiers for relationships, which permits the same seed logic to work across environments with different internal IDs.

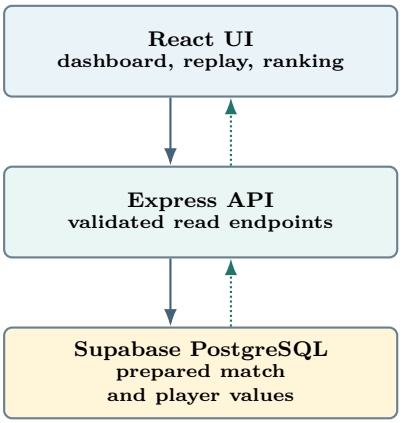


Fig. 2. Runtime request path. Requests travel from the React interface through the Express API to Supabase PostgreSQL. Responses are returned from the database to the Express API and then forwarded to the React interface.

The database-check procedure verifies the required tables, row-level security configuration, permissions, coordinate limits, and finite VAEP values after import. It also checks that the 172 regular-play goals have possession IDs, no shootout goals are labeled, and the credited goal totals agree with each match's final score. The audit found zero goal-score, VAEP component-sum, or player-stat aggregation mismatches. These are internal consistency checks: they do not independently reproduce every feature, label, probability, or tactical interpretation.

The reproducibility artifacts include

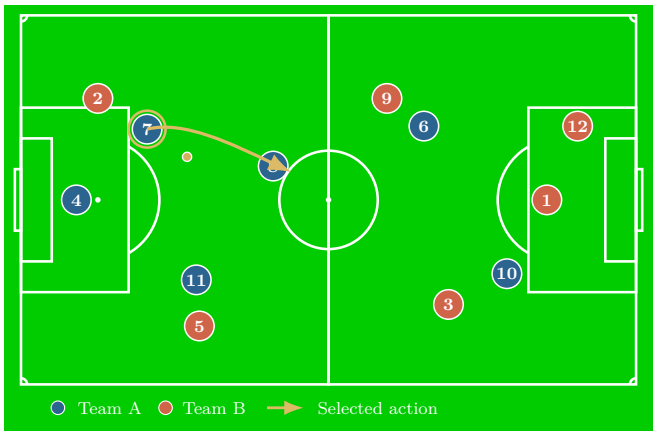


Fig. 3. Conceptual action-replay view. The field geometry includes the penalty areas, goal areas, penalty spots, penalty arcs, centre line, and corner arcs. Player markers and the highlighted trajectory illustrate how a stored SPADL action can be presented for interactive inspection.

`pipeline/run_vaep.py` for seed generation, `pipeline/validate_vaep.py` for match-grouped validation, and `pipeline/enrich_goal_stories.py` for goal-story enrichment, together with `data/validation_report.txt` and `data/paper_validation_addendum.txt`. StatsBomb source files and generated seed/model artifacts are local project inputs and are not embedded in this manuscript; reproduction requires obtaining the open data and complying with its access and attribution terms.

V. Discussion

VAEP values are computed during offline processing and stored before the application is deployed. At runtime, the browser retrieves the prepared values through the API, while model training and VAEP calculation remain outside the request path. This also keeps database credentials away from the client and limits the browser to data access and visualization.

Because intermediate records are retained, an action value can be traced to its source event, generated features, and model output. This is relevant when actions with similar visible outcomes occur in different game states.

The validation results do not establish predictive superiority or operational suitability. Although ROC AUC exceeds 0.5 for both tasks, the fixed 0.50 decision threshold detects few scoring positives and no conceding positives. High accuracy is dominated by class imbalance. The five match-level holdouts overlap, cover only one tournament, and were not used for hyperparameter or threshold selection. Model calibration and external-season generalization remain unestablished. Stored-seed integrity checks demonstrate internal consistency, not that every feature, label, or action value is substantively correct. Goal buildup is likewise descriptive event-feed context and does not establish causality.

The goal-story enrichment is separated from the model stage: it adds regular-play goal identity, credited side, score-after, and possession metadata while preserving the previously calculated VAEP values. This supports a fan-oriented replay that starts with the score-changing event and its recorded possession, while keeping VAEP as context rather than treating a high score as proof of a goal.

VI. Conclusion

*VAEP Explorer*connects football event data, contextual action valuation, persistent storage, and interactive analysis in a single workflow. StatsBomb event files are standardized through SPADL, contextual features and short-horizon labels are generated, two CatBoost classifiers estimate scoring and conceding probabilities, and their state-to-state changes are combined into VAEP. A separate enrichment step adds goal and possession context without recalculating VAEP.

The verified dataset contains all 64 FIFA World Cup 2022 matches, 680 players, 138,946 actions, 172 reconciled regular-play goals, and 680 player-stat records. It is stored in Supabase PostgreSQL, exposed through an Express API, and rendered through a React analytical interface.

Five repeated 48/16 match-grouped holdouts compare fresh XGBoost and CatBoost fits. They reveal useful ranking signal but weak threshold-based rare-event detection, especially for conceding outcomes. The system therefore contributes an auditable, exploratory analysis workflow; it does not establish a production-ready predictor or cross-competition generalization.

VII. Future Work / Recommendations

Future evaluation should use a predeclared non-overlapping match-level test set or external seasons and competitions, with sufficient positive examples for stable rare-event assessment. Thresholds should be selected only on training/validation data and reported with precision–recall trade-offs. Calibration should be evaluated explicitly, and feature, label, and stored-probability provenance should be versioned alongside model artifacts. Independent domain review is needed to assess whether action valuations are tactically plausible; arithmetic consistency alone cannot provide that judgment.

On the application side, production deployment should add restrictive CORS configuration, rate limiting, automated integration tests, and virtualized or paginated replay rendering for larger action timelines. These changes would support larger datasets and longitudinal action-value analysis.

References

[1] T. Decroos, L. Bransen, J. Van Haaren, and J. Davis, "Actions Speak Louder than Goals: Valuing Player Actions in Soccer," in *Proc. 25th ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, Anchorage, AK, USA, 2019, pp. 1851–1861, doi: 10.1145/3292500.3330758.

[2] L. Pappalardo, P. Cintia, A. Rossi, E. Massucco, P. Ferragina, D. Pedreschi, and F. Giannotti, "A public data set of spatio-temporal match events in soccer competitions," *Sci. Data*, vol. 6, art. no. 236, 2019, doi: 10.1038/s41597-019-0247-7.

[3] S. Giancola, M. Amine, T. Dghaily, and B. Ghanem, "SoccerNet: A scalable dataset for action spotting in soccer videos," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition Workshops*, Salt Lake City, UT, USA, 2018, pp. 1711–1721, doi: 10.1109/CVPRW.2018.00223.

[4] J. Fernández, L. Bornn, and D. Cervone, "A framework for the fine-grained evaluation of the instantaneous expected value of soccer possessions," *Mach. Learn.*, vol. 110, pp. 1389–1427, 2021, doi: 10.1007/s10994-021-05989-6.

[5] S. Le Coz, F. Boustila, and F. Imbach, "A semi-Markov framework for modeling football possessions and temporal expected threat," *Sci. Rep.*, vol. 16, art. no. 22590, 2026, doi: 10.1038/s41598-026-52938-1.

[6] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, San Francisco, CA, USA, 2016, pp. 785–794, doi: 10.1145/2939672.2939785.

[7] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, "CatBoost: Unbiased boosting with categorical features," in *Proc. 32nd Int. Conf. Neural Information Processing Systems*, 2018, pp. 6639–6649.

[8] StatsBomb, "StatsBomb Open Data," GitHub repository. [Online]. Available: https://github.com/statsbomb/open-data. Accessed: Sep. 28, 2026.

[9] ML-KU Leuven, "socceraction documentation: SPADL and VAEP." [Online]. Available: https://socceraction.readthedocs.io/en/stable/. Accessed: Sep. 28, 2026.

[10] H. Janetzko, D. Sacha, M. Stein, T. Schreck, D. A. Keim, and O. Deussen, "Feature-driven visual analytics of soccer data," in *Proc. IEEE Conf. Visual Analytics Science and Technology*, Paris, France, 2014, pp. 13–22, doi: 10.1109/VAST.2014.7042477.

[11] E. Delibas, A. Uzun, M. F. Inan, O. Guzey, and A. Cakmak, "Interactive exploratory soccer data analytics," *INFOR*, vol. 57, no. 2, pp. 141–164, 2019, doi: 10.1080/03155986.2018.1533204.